

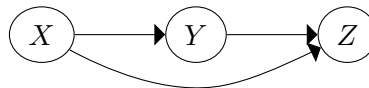
# Topic 17: Graphical Representations of Random Variables

02-680: Essentials of Mathematics and Statistics

September 4, 2026

## 1 Bayesian Networks

Sometimes we need a more complicated conditional description. One way to do this is using a network (graph).



In general for a network such as this we define the probabilities as:

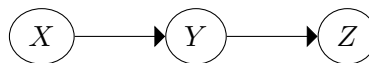
$$p(\langle Q_1, Q_2, \dots, Q_d \rangle) = \prod_{i=1}^d p\left(Q_i \mid \bigwedge_{j \in N^{in}(Q_i)} Q_j\right)$$

remember that  $N^{in}(v)$  is the set of in-neighbors of  $v$  in a graph.

So for the Bayesian Network above, we may have something like:

| $p(X = x)$ |     | $p(Y \mid X)$ |         | $p(Z \mid Y, X)$ |         |
|------------|-----|---------------|---------|------------------|---------|
| $x$        |     | $x$           | $y = 1$ | $y = 0$          |         |
| 0          | 0.7 | 0             | 0.5     | 0.5              | 0       |
| 1          | 0.3 | 1             | 0.9     | 0.1              | 1       |
|            |     |               |         |                  | $z = 1$ |
|            |     |               |         |                  | $z = 0$ |
|            |     |               |         |                  | 0       |
|            |     |               |         |                  | 1       |
|            |     |               |         |                  | 0       |
|            |     |               |         |                  | 1       |

But if we alter the network to the following (i.e. remove the dependence of  $Z$  on  $X$ ),



we end up with

| $x$ | $p(X = x)$ | $p(Y   X)$ |         |     | $p(Z   Y)$ |         |         |
|-----|------------|------------|---------|-----|------------|---------|---------|
| $x$ |            | $y = 1$    | $y = 0$ |     | $y$        | $z = 1$ | $z = 0$ |
| 0   | 0.7        | 0          | 0.5     | 0.5 | 0          | 0.2     | 0.8     |
| 1   | 0.3        | 1          | 0.9     | 0.1 | 1          | 0.7     | 0.3     |

## 2 Markov Chains

Markov chains are usually used to model sequences of items. We saw when talking about conditional probabilities that:

$$p(X_n, X_{n-1}, \dots, X_1) = p(X_n | X_{n-1}, \dots, X_1) p(X_{n-1}, \dots, X_1) \quad (1)$$

$$= p(X_n | X_{n-1}, \dots, X_1) p(X_{n-1} | X_{n-2}, \dots, X_1) p(X_{n-2}, \dots, X_1) \quad (2)$$

$$\dots \quad (3)$$

$$= p(X_n | X_{n-1}, \dots, X_1) \dots p(X_2 | X_1) p(X_1) \quad (4)$$

$$(5)$$

The **Markov assumption** is that the probability of  $X_k$  depends on  $X_{k-1}$  alone. This simplifies the computation, but at the cost of long term connections. Anecdotally the Markov assumption says:

the future is independent of the past given the present.

So that means we can rewrite the statement above as:

$$p(X_n, X_{n-1}, \dots, X_1) = p(X_n | X_{n-1}) p(X_{n-1} | X_{n-2}) \dots p(X_2 | X_1) p(X_1).$$

This is the basis for **Hidden Markov Models** which are used a lot in genetics.

**Example.** Consider a sequence of DNA nucleotides modeled as a set of random variables  $N_1 N_2 \dots N_n$ , each of which can take on the value  $\{A, T, C, G\}$ . We want to know if the sequence is a CpG island or not.

Looking at the equation above, if we know the initial distribution (that is the probability distribution for  $p(N_1)$ ) for both cases, and the transition distributions ( $p(N_k | N_{k-1})$ ) we can determine which is more probable.

$$P_b = \begin{bmatrix} & A & C & G & T \\ A & 0.30 & 0.20 & 0.29 & 0.21 \\ C & 0.32 & 0.30 & 0.08 & 0.30 \\ G & 0.25 & 0.25 & 0.29 & 0.21 \\ T & 0.18 & 0.24 & 0.29 & 0.29 \end{bmatrix} \quad P_c = \begin{bmatrix} & A & C & G & T \\ A & 0.18 & 0.27 & 0.43 & 0.12 \\ C & 0.17 & 0.37 & 0.27 & 0.19 \\ G & 0.16 & 0.34 & 0.37 & 0.13 \\ T & 0.08 & 0.36 & 0.38 & 0.18 \end{bmatrix}$$

Assuming for both  $p(N_1 = \mathbf{A}) = p(N_1 = \mathbf{C}) = p(N_1 = \mathbf{T}) = p(N_1 = \mathbf{G}) = 0.25$ , what is the sequence **ACTTC** more likely to be a CpG island or not?

## Useful References

Degroot and Schervish. “Probability and Statistics” §§3.10